

2. Requirement Analysis

Introduction

Requirement Analysis is one of the most important phases in software development. In this phase, all system requirements are identified, analyzed, and documented. The purpose of this phase is to understand what functionalities the system should provide and how the system should perform under different conditions.

For the AI-Powered Smart Visitor Management System, the requirements were gathered based on the needs of organizations that require secure and efficient visitor management.

Functional Requirements

Functional requirements describe the features and operations that the system must perform.

Visitor Registration

The system allows users to register visitors by entering visitor details through a digital form.

Information Captured

- Visitor Name
- Mobile Number
- Email Address
- Purpose of Visit
- Visit Date

Benefits

- Easy visitor registration
 - Faster data entry
 - Accurate visitor information
-

AI Recommendation

The system automatically generates recommendations based on the visitor's purpose.

Examples

Purpose	AI Recommendation
Interview	Schedule HR Meeting Room
Client Meeting	Arrange Conference Room

Purpose	AI Recommendation
Delivery	Route to Security Desk
Training	Allocate Training Hall

Benefits

- Intelligent visitor handling
 - Reduced manual decision making
 - Improved operational efficiency
-

Email Notification

The system sends automatic email notifications when a visitor request is submitted.

Email Contains

- Visitor Name
- Purpose
- Visit Date
- Status

Benefits

- Instant communication
 - Better visitor awareness
 - Reduced manual follow-up
-

Check-In and Check-Out

Security personnel can track visitor movement using Check-In and Check-Out buttons.

Check-In Process

Updates status to:

Checked In

Check-Out Process

Updates status to:

Checked Out

Benefits

- Real-time visitor tracking
 - Improved security monitoring
 - Accurate visitor records
-

Reports and Dashboard

The system automatically generates reports and graphical dashboards.

Available Reports

- Visitor Status Report
- Visitor Purpose Report
- Monthly Visitor Report

Dashboard Features

- Pie Charts
- Bar Charts
- Monthly Trends

Benefits

- Better decision making
 - Real-time analytics
 - Improved monitoring
-

Non-Functional Requirements

Non-functional requirements define how the system should perform.

Security

The system should ensure that only authorized users can access visitor information.

Security Features

- User Authentication
- Role-Based Access
- Secure Data Storage

Benefits

- Data protection
 - Improved confidentiality
 - Secure access control
-

Reliability

The application should work consistently without failures.

Requirements

- Accurate data storage
- Reliable workflow execution
- Consistent performance

Benefits

- Reduced system errors
 - Better user experience
-

Performance

The system should respond quickly to user actions.

Performance Goals

- Fast form submission
- Quick dashboard loading
- Immediate validation responses

Benefits

- Improved productivity
 - Better user satisfaction
-

Scalability

The system should support future growth.

Scalability Features

- Large visitor database support
- Additional modules can be added
- Future integration capabilities

Benefits

- Long-term usability
 - Easy expansion
-

User Roles

The application supports different user roles.

Admin

Responsibilities

- Manage application settings
- View reports and dashboards
- Monitor visitor activities
- Maintain system configurations

Permissions

- Full access to the application
-

Employee

Responsibilities

- Create visitor requests
- View submitted requests
- Track visitor status

Permissions

- Create and view visitor records
-

Security Personnel

Responsibilities

- Verify visitor information
- Perform Check-In
- Perform Check-Out

Permissions

- Update visitor status
-

Software Requirements

The following software components are required.

ServiceNow Developer Instance

Used for application development and deployment.

Web Browser

Supported Browsers:

- Google Chrome
 - Microsoft Edge
 - Mozilla Firefox
-

Internet Connection

Required for accessing the ServiceNow cloud platform.

Programming Technologies

- JavaScript
 - Glide API
 - Flow Designer
 - Business Rules
 - Client Scripts
-

Hardware Requirements

The following hardware resources are recommended.

Laptop/Desktop

Minimum Specifications:

- Intel Core i3 Processor or above
 - 4 GB RAM
 - 20 GB Free Storage
-

Network Connectivity

Stable internet connection for uninterrupted access.

Display

Minimum resolution:

1366 x 768

or higher.

Feasibility Study

The feasibility study determines whether the project can be successfully developed and implemented.

Technical Feasibility

The project is technically feasible because ServiceNow provides all required tools and modules.

Available Technologies

- App Engine Studio
- Flow Designer
- Business Rules
- Client Scripts
- Reports
- Dashboards

Conclusion

The project can be developed successfully using ServiceNow.

Economic Feasibility

The project is economically feasible because no additional software licenses are required.

Cost Factors

- ServiceNow Developer Instance: Free
- Development Tools: Free
- Open Internet Access

Conclusion

The project can be implemented at minimal cost.

Operational Feasibility

The system is easy to use and maintain.

Advantages

- User-friendly interface
- Simple workflows
- Automated notifications
- Easy report generation

Conclusion

The system can be adopted easily by organizations.

Legal Feasibility

The system stores visitor information securely and follows standard data management practices.

Benefits

- Secure data storage
 - Controlled access
 - Data privacy support
-

Summary of Requirements

Requirement Type	Description
Functional	Visitor Registration, AI Recommendation, Email Notification, Check-In/Check-Out, Reports
Non-Functional	Security, Reliability, Performance, Scalability
Software	ServiceNow, Browser, Internet
Hardware	Laptop/Desktop, RAM, Storage
Feasibility	Technical, Economic, Operational, Legal

Conclusion

The Requirement Analysis Phase identified all functional and non-functional requirements necessary for developing the AI-Powered Smart Visitor Management System. The analysis confirms that the project is

technically feasible, economically affordable, and operationally effective. These requirements serve as the foundation for the design and development phases of the project.